Cal/Val of P-band BIOMASS aboveground biomass in representative Southeast Asia forests using field and lidar data

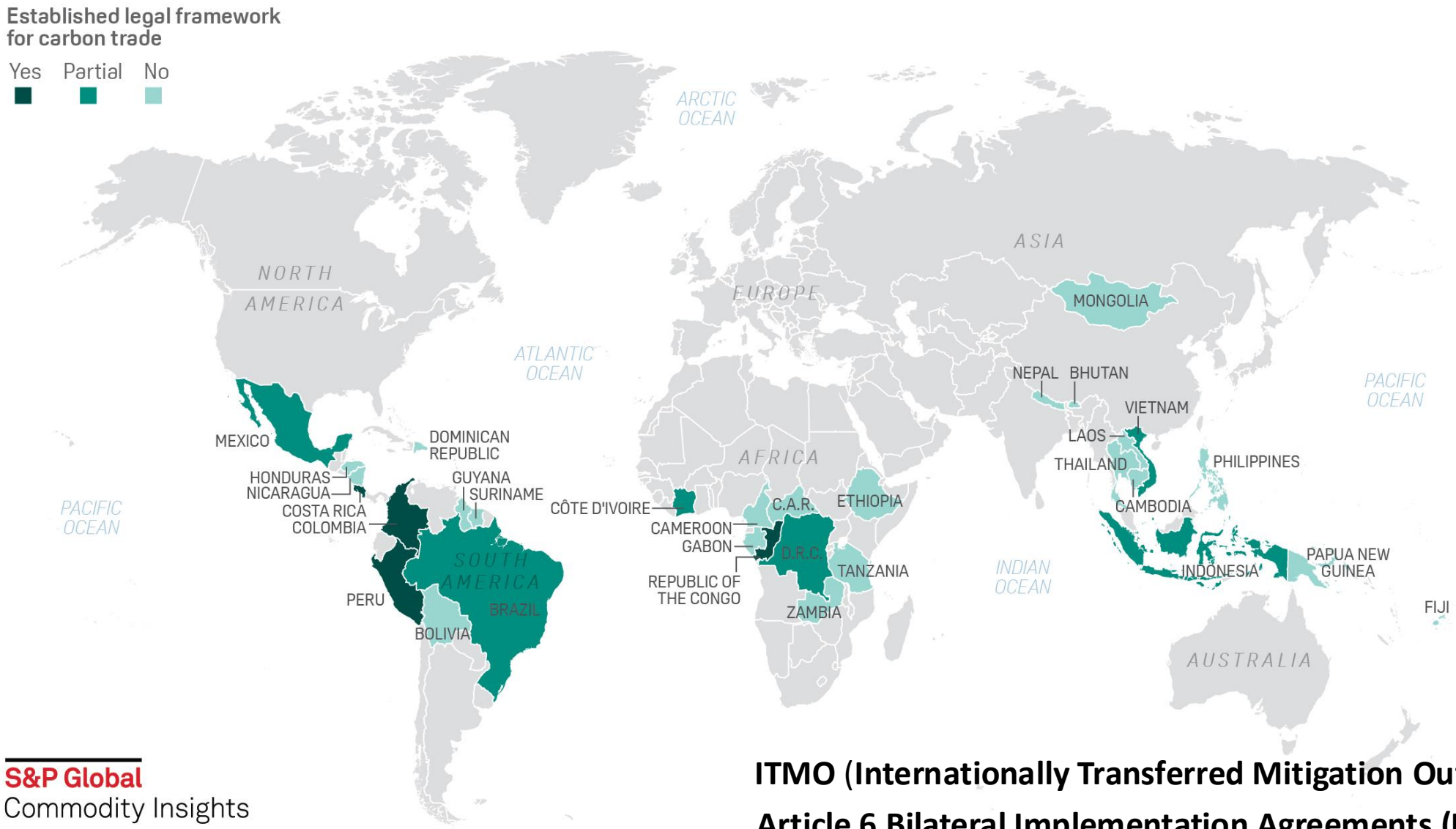
PI: Hao Tang, Lee Ken Yoong, Kim Hwa Lim
Collaborator: Soo Chin Liew, Le Toan Thuy

Department of Geography
Centre for Remote Imaging, Sensing and Processing
National University of Singapore

22-Oct-2026

COUNTRIES WITH LEGAL FRAMEWORK TO SUPPORT CARBON-LINKED TRANSACTIONS

Out of 31 countries included in the Rights and Resources Initiative's August 2021 technical report, only four have established a legal framework for the trade of carbon credits. All highlighted countries, except the Dominican Republic, have laws for securing community tenure to forests.



S&P Global
Commodity Insights

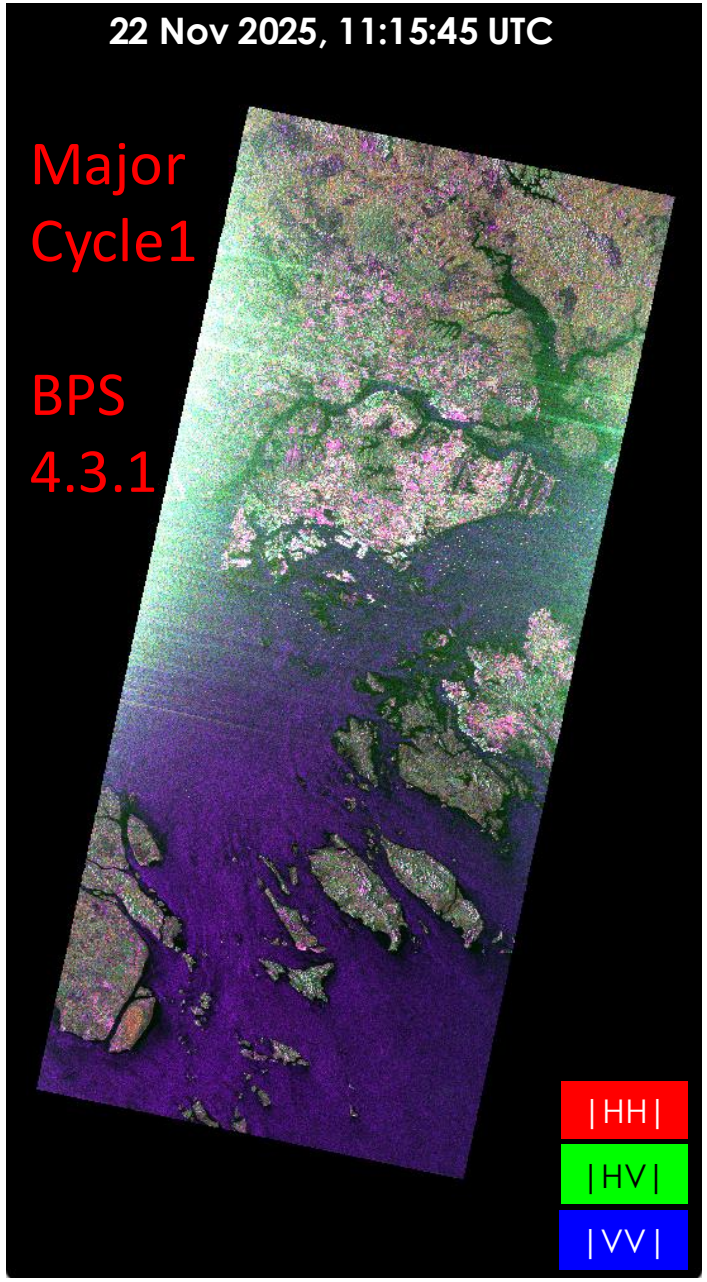
Source: S&P Global Commodity Insights, Rights and Resources Initiative

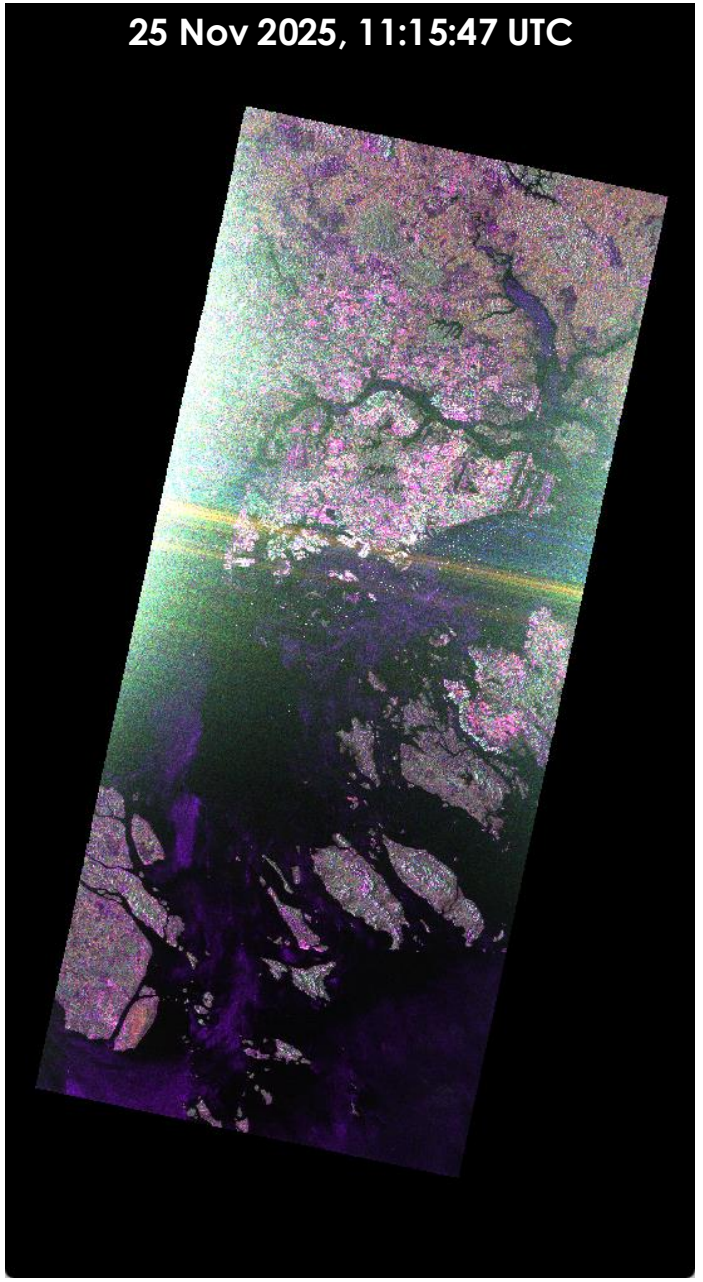
ITMO (Internationally Transferred Mitigation Outcome)

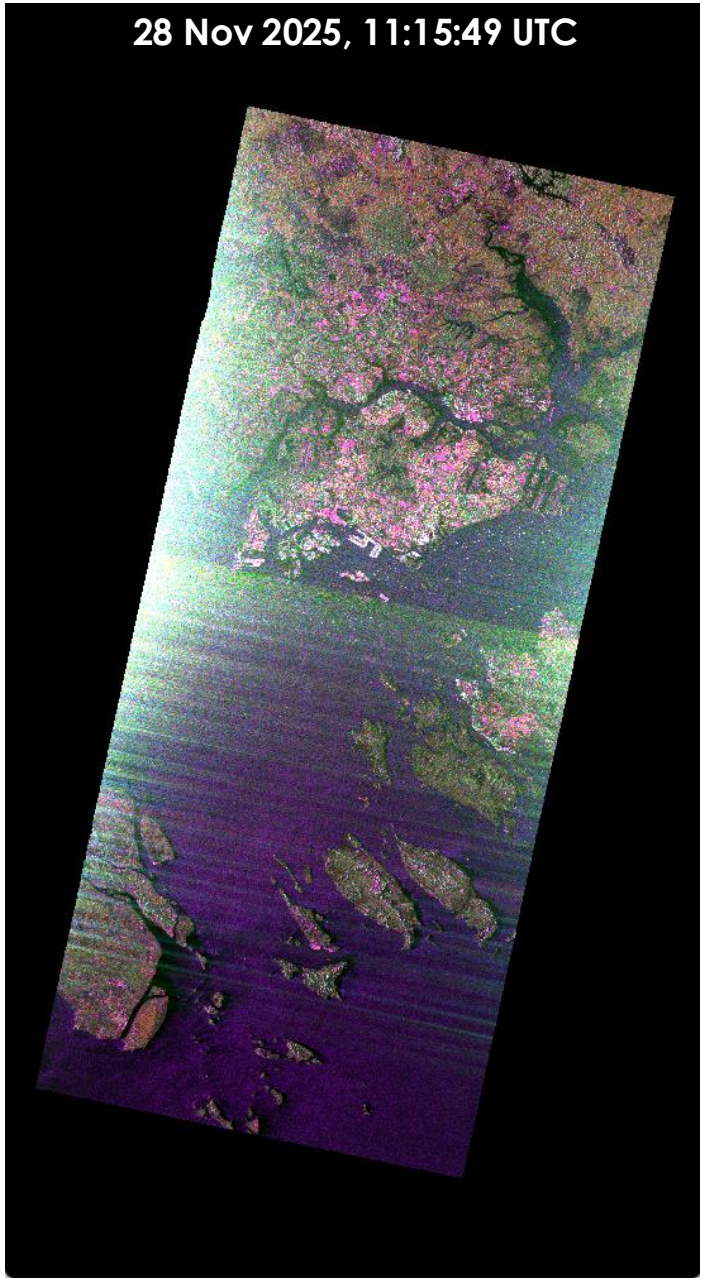
Article 6 Bilateral Implementation Agreements (IAs):

Singapore actively builds sovereign carbon credit pipelines under Article 6 of the Paris Agreement

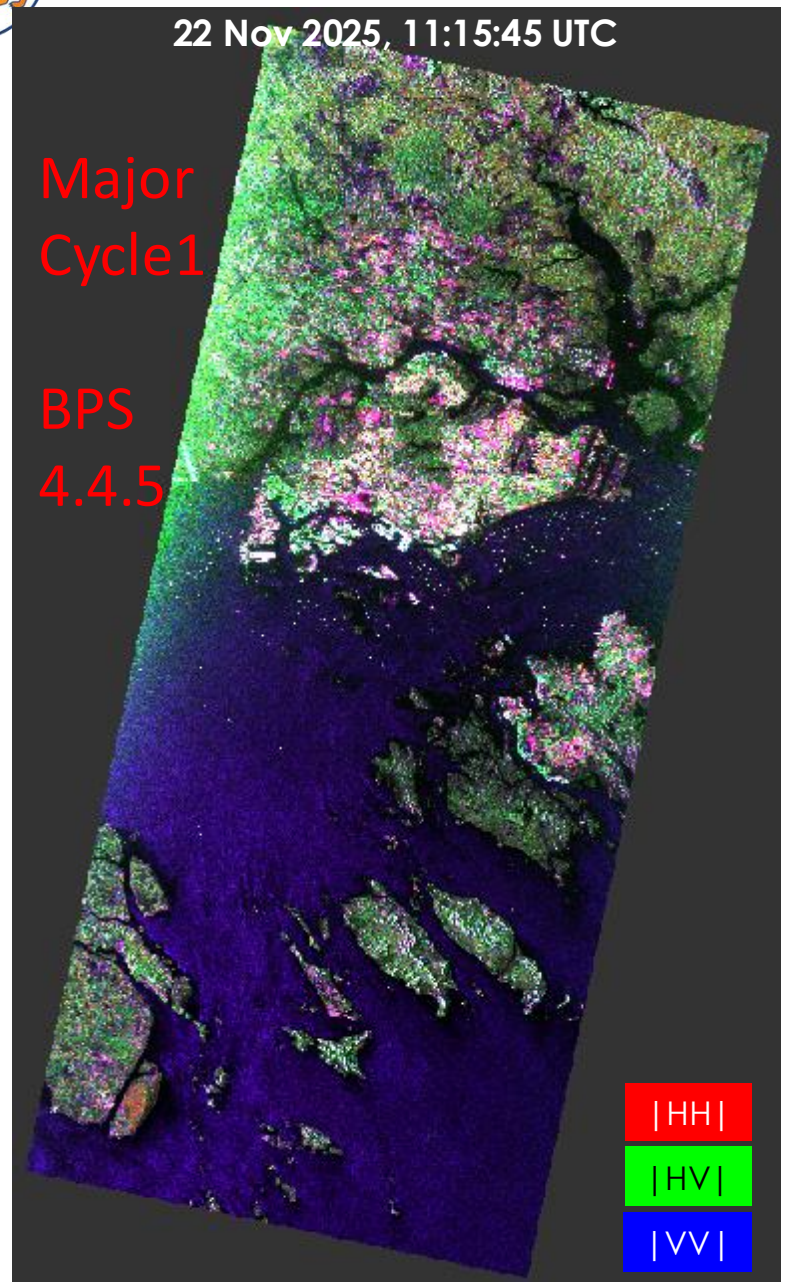
Biomass Orthorectified Scenes Over Singapore

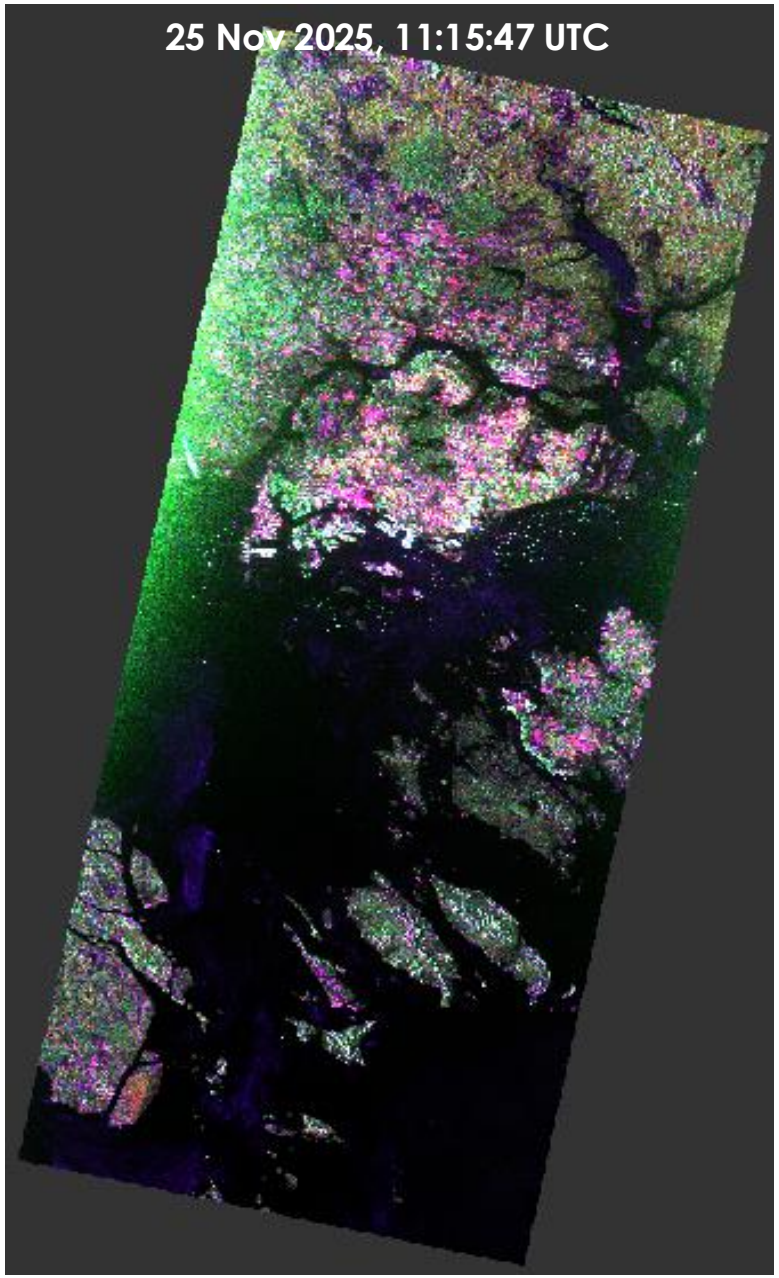


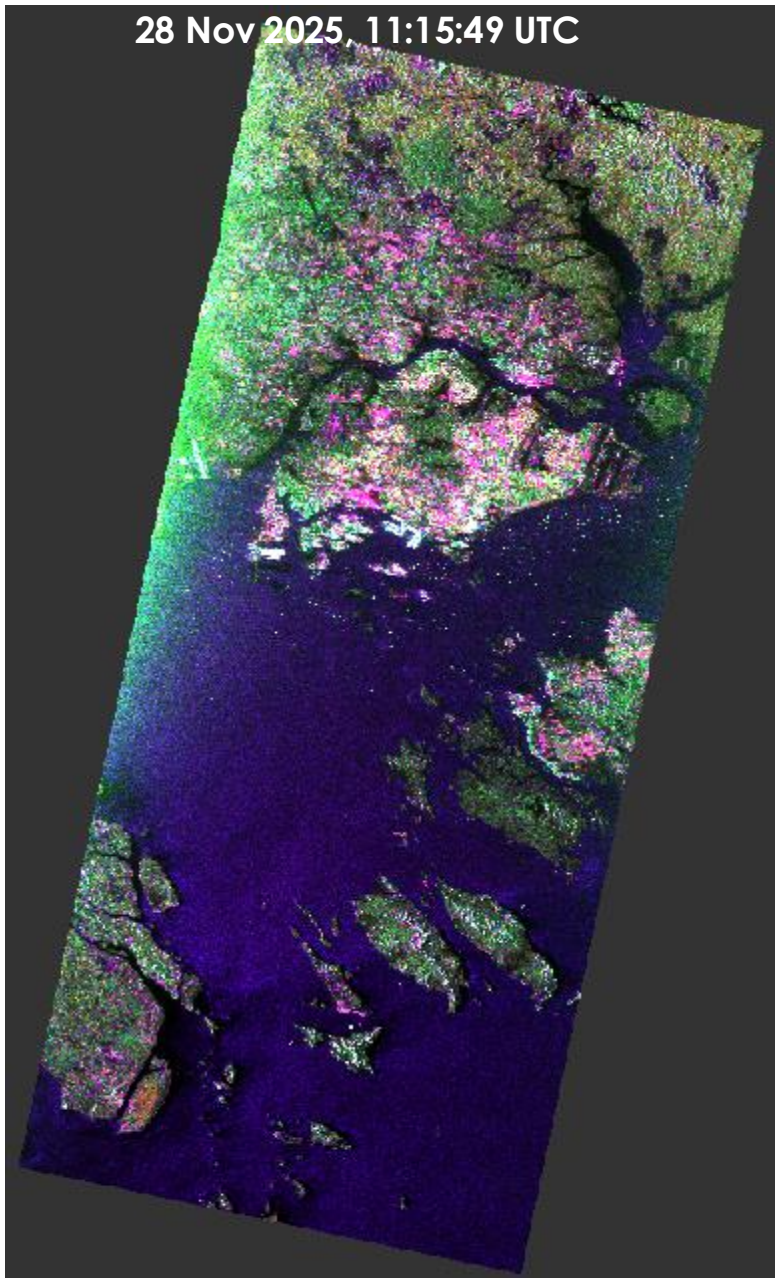




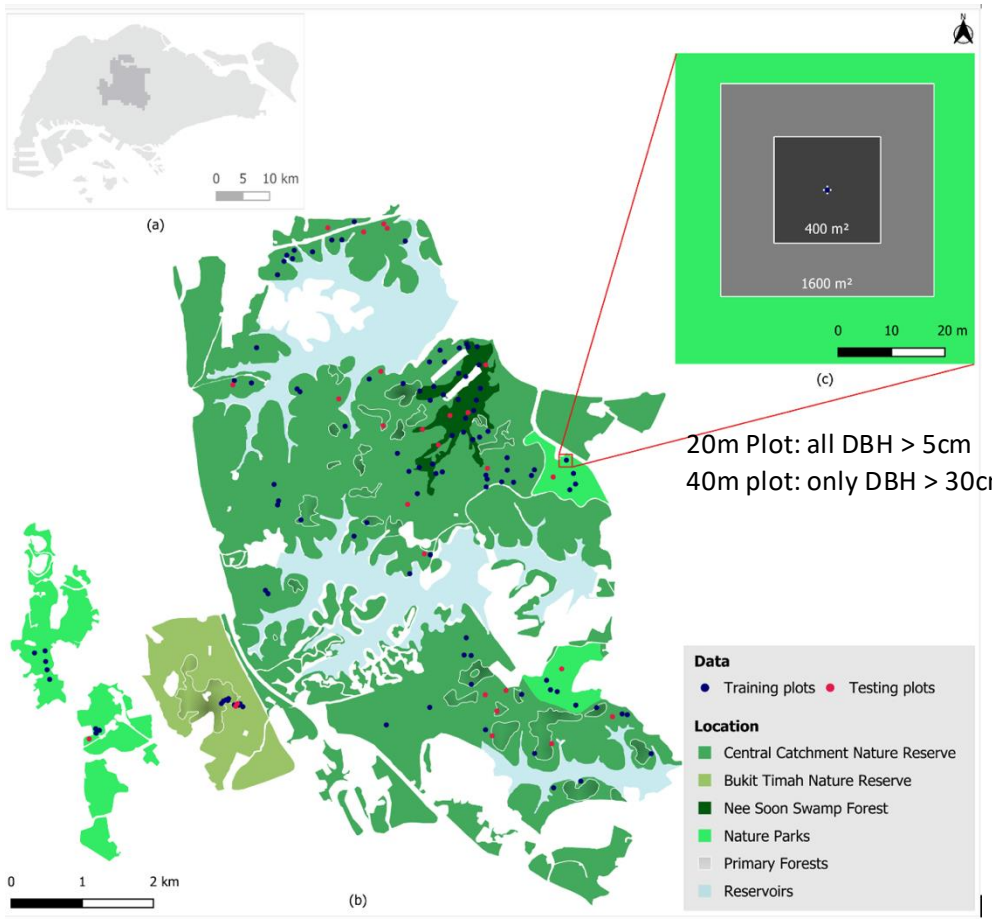
Biomass Orthorectified Scenes Over Singapore







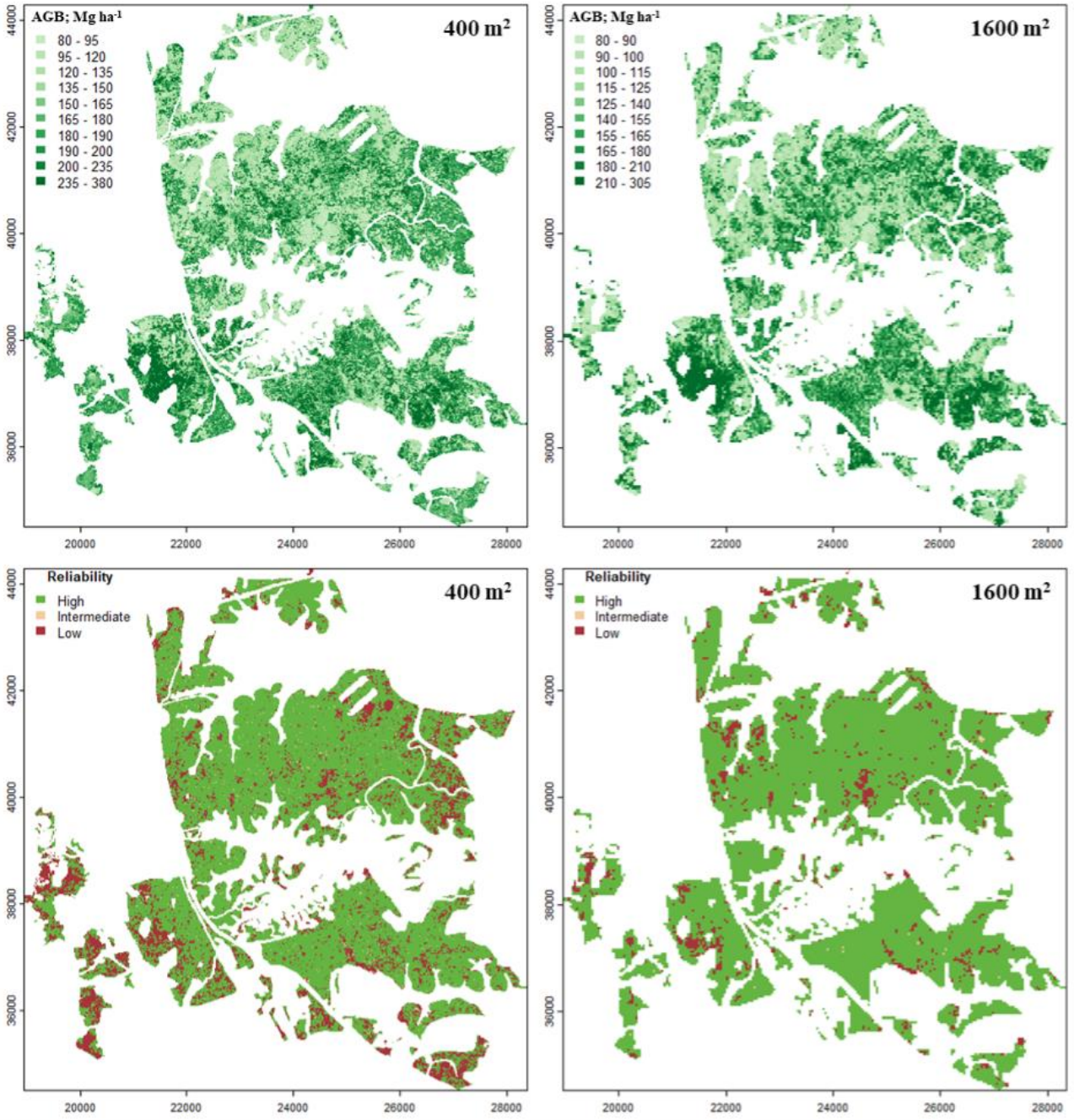
Primary reference map (Field + ALS)



Field inventory (N = 112; 2022~2024) + Airborne lidar* (2024)

Forest attribute	Plot size (m^2)	Prediction mean			RMSE%		
		Cal	Val	Test	Cal	Val	Test
AGB (Mg/ha)	20 X 20m	179.9	178.9	173.4	29.8	41.9	47.5
	40 X 40m	152.5	151.5	153.7	28.2	36.5	39.5

*Leica CityMapper-2, > 30 pts/ m2

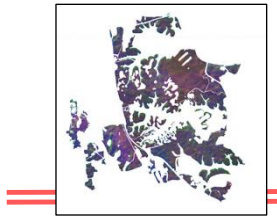


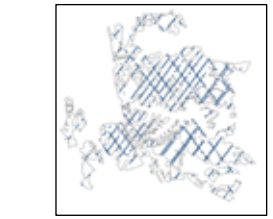
Secondary reference map (GEDI interpolation and uncertainty)

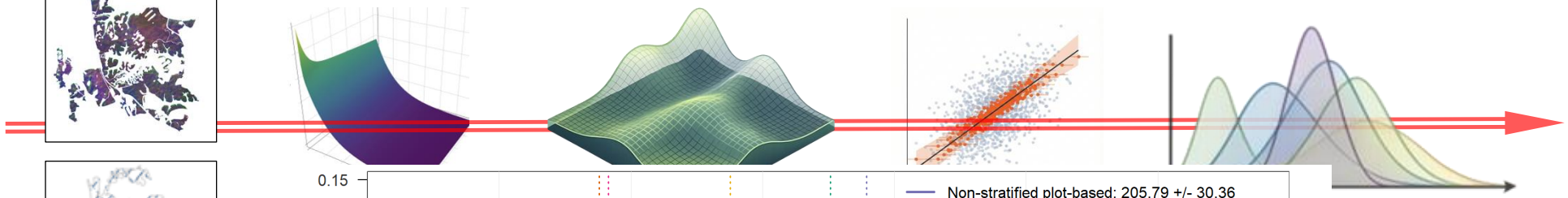
Input

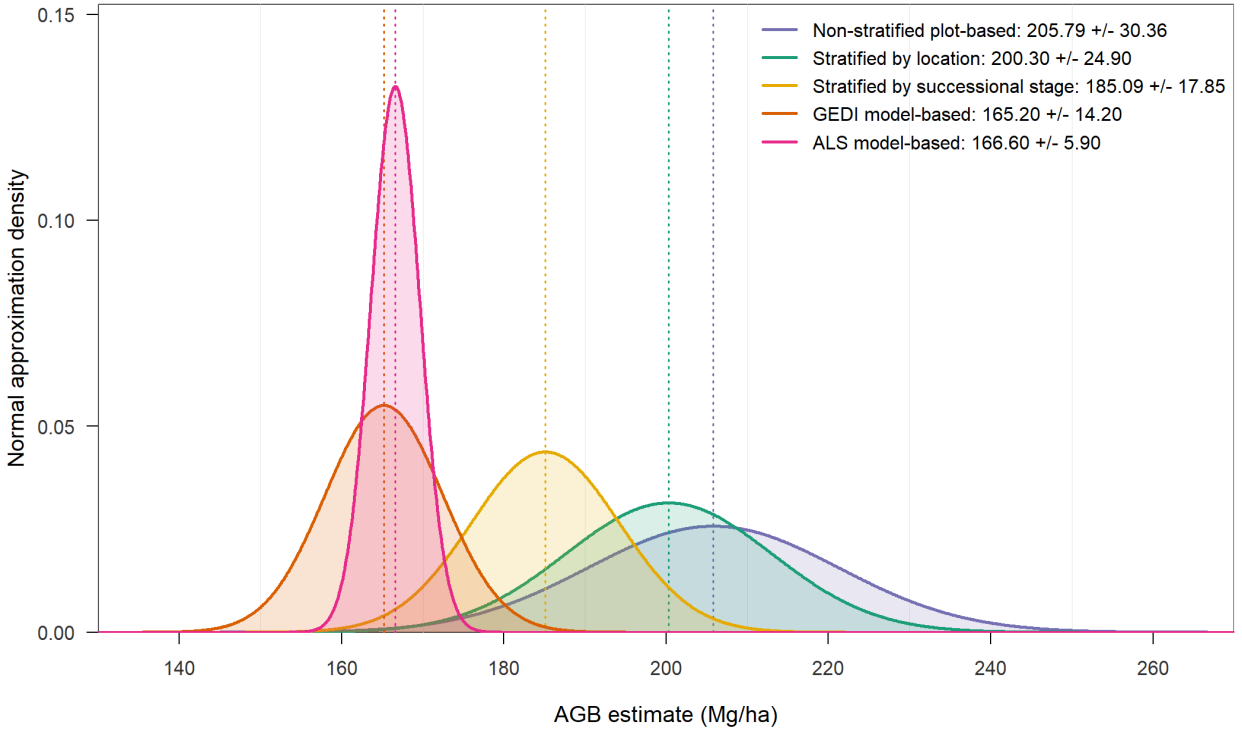
Interpolated RH

Error propagation









Pixel-level predictions with quantified uncertainty





Model-based estimator

Interpolation-based mean AGBD estimate: 165.2 ± 14.2 Mg ha^{-1}

Airborne Lidar model-based AGBD estimate: 166.6 ± 5.9 Mg ha^{-1}

(Note the two have completely different model assumptions; thus not directly comparable)

Terrestrial Laser Scanning (TLS) + Scatterometer Experiment

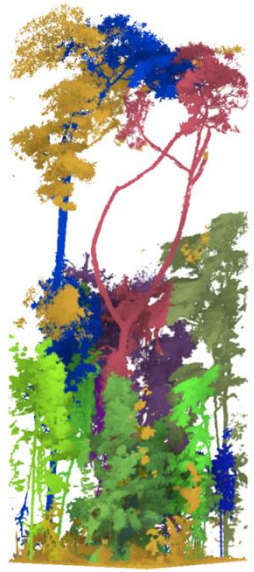
Improve allometric equation (~30 plots)
Test BIOMASS sensitivity towards complex canopy structure



3DFin
(open-source code)



LIS Pro 3D
(commercial software)



Individual Tree QSM



Benchmark TropiScat and AfriScat experiment to explore temporal coherence



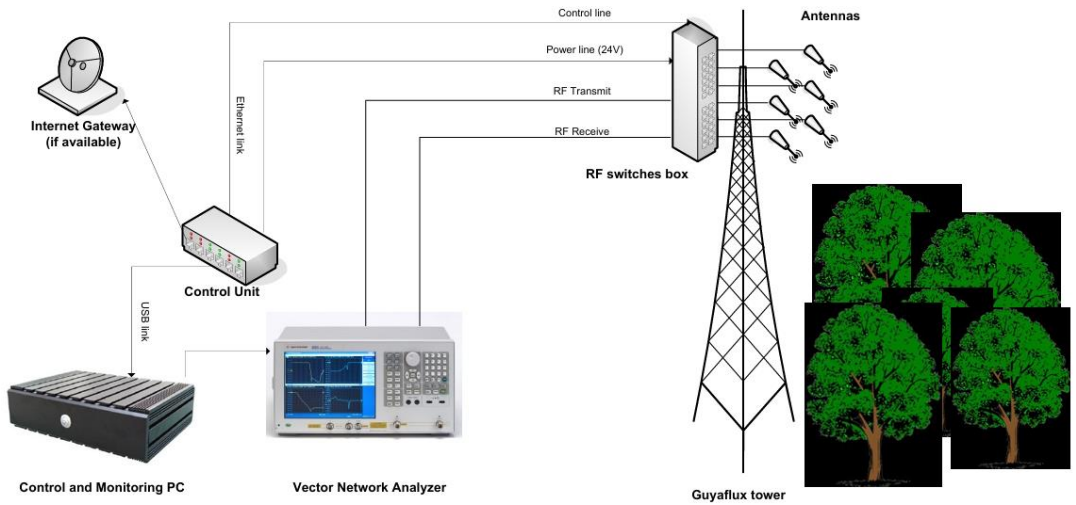
Keysight E5063A (15kg)
~ 43.2 x 29.7 x 23.5 cm

Frequency range : 100 kHz to 3 GHz

Frequency resolution : 1 Hz
Typical dynamic range : 122 dB for 100 MHz to 4.35 GHz
Upgradable to 18 GHz

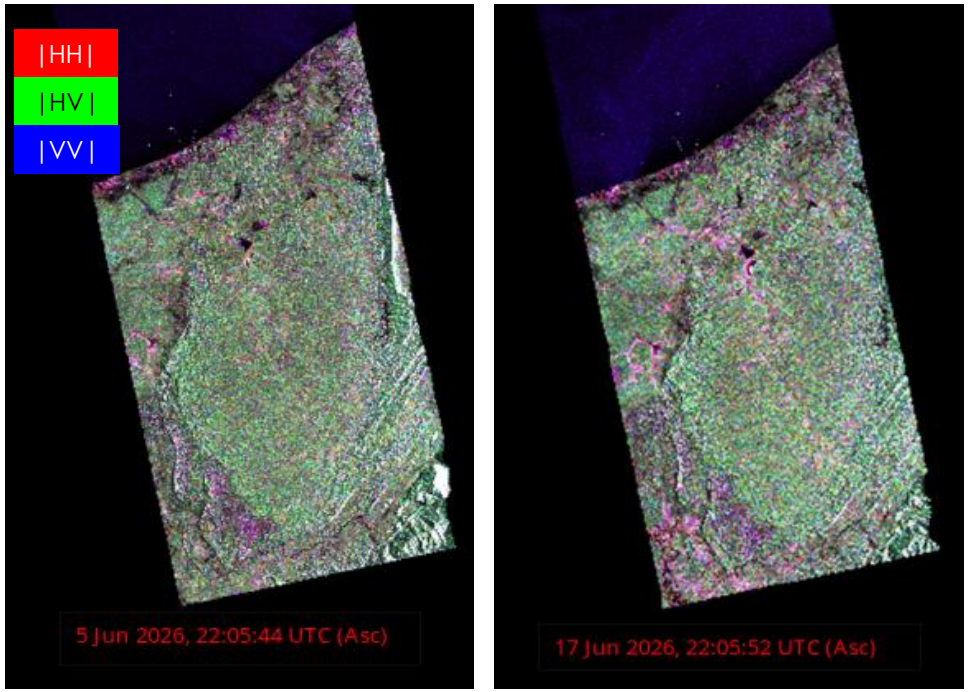
Operating temperature range : 5 to 40 °C

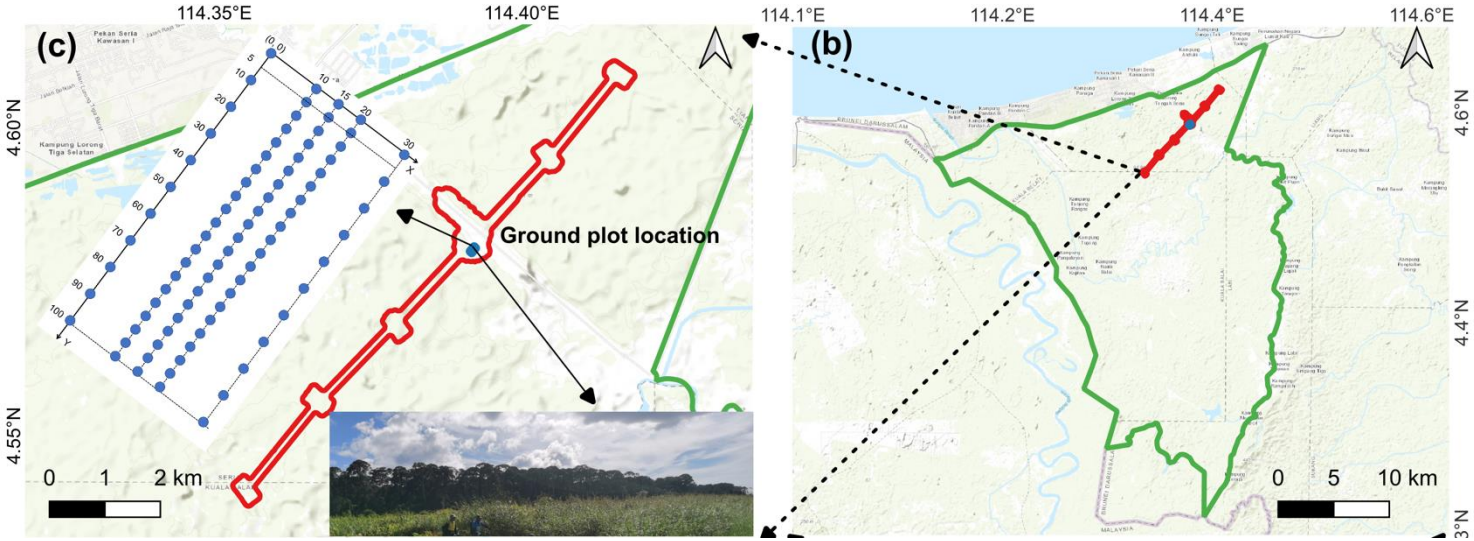
Power range : -20 to 0 dBm (300 kHz to 8.5 GHz)

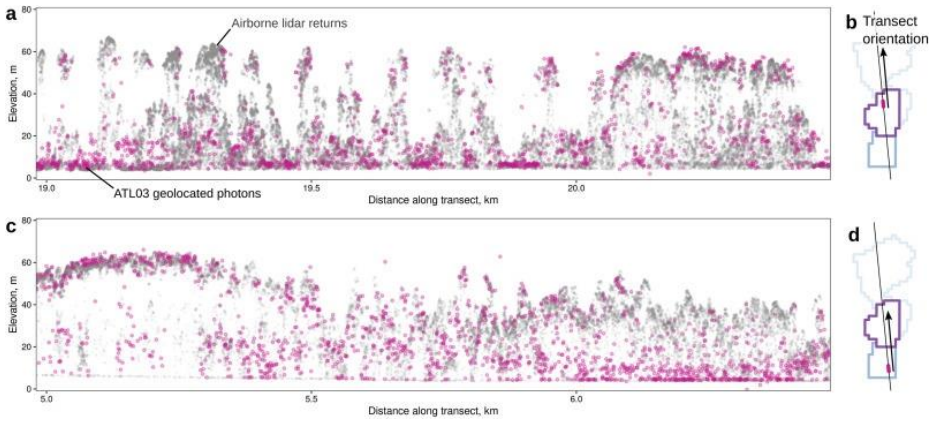


Alternative site(s)

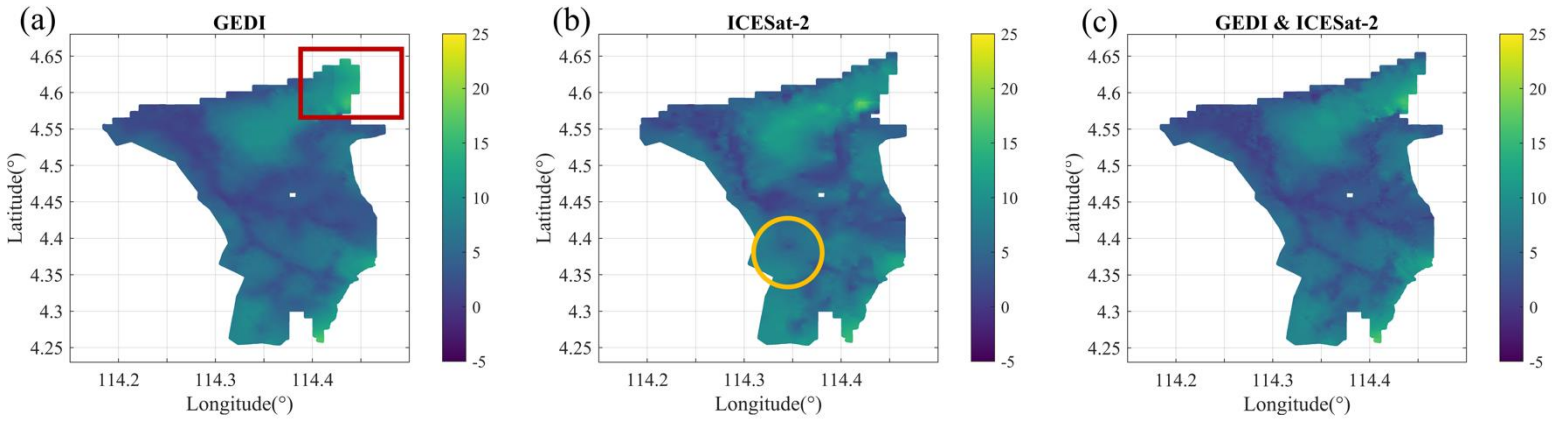
intact peatland forest in Brunei







Cobb et al. (2023)



Sun et al. (2026)

Acknowledgements

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